4 Fig. 4.1 shows a pan with a copper base on a hotplate. The hotplate heats the pan and the water.

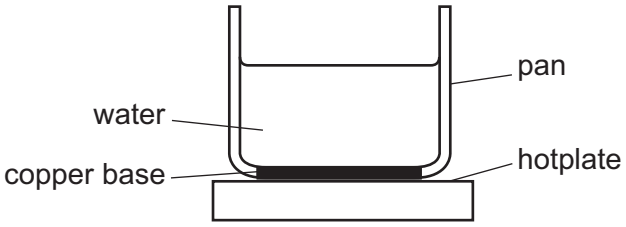

Fig. 4.1

(a) Explain how thermal energy is conducted through the copper base.

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.....

..... [3]

(b) **(i)** Define, in words, specific heat capacity.

.....

.....

..... [2]

(ii) A student heats a metal object to 100 °C. The student places the metal object in an insulated cup containing water at 22 °C. The final temperature of the water and the metal object is 31 °C.

The specific heat capacity of water is 4.2 J / (g °C).
The mass of the water in the insulated cup is 50 g.
The mass of the metal object is 54 g.

Calculate the specific heat capacity of the metal.

specific heat capacity = ... [3]

[Total: 8]

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